Meat, fish, liver are rich sources of iron. Cereals, nuts, spinach, dates are good plant sources of iron.

Absorption and excretion

Normally, about 5 to 10% of dietary iron is absorbed by the active transport process. Most absorption occurs in the duodenum. Infants and children absorb a higher percentage of iron from food than adults. Iron deficient children absorb twice as much as that of normal children. Excess dietary iron is stored as ferritin. Only lesser amounts are excreted in the urine, feces and sweat.

Factors affecting iron absorption

1. Impaired iron absorption takes place in patients who have total removal of stomach or a removal of the considerable amount of the intestine.

2. A diet high in phosphate causes decreased absorption due to the formation of insoluble ferric phosphate.

3. Copper deficiency reduces iron absorption, as copper helps in transport of iron.

4. Phytic acid and oxalic acid interfere with iron absorption.

5. Vitamin C increases iron absorption.

Deficiency

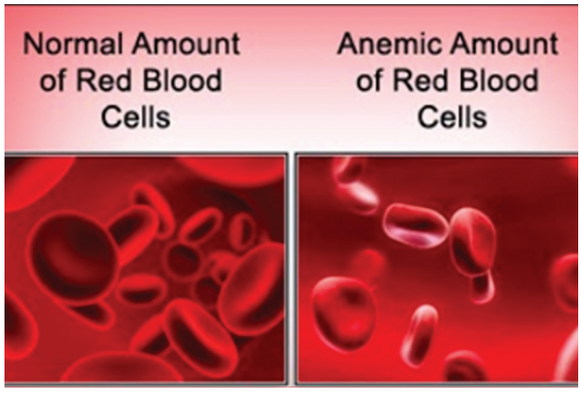


Figure 9.12 Iron deficiency anemia

Deficiency of iron causes iron deficiency anemia. The deficiency may be due to dietary or lack of iron absorption due to gastro intestinal tract diseases like diarrhoea.

9.3.2. Copper

Copper is an essential micro element. Copper content in an adult human is about 100 mg. Copper in whole blood is almost evenly distributed between cell and plasma. The highest concentrations are found in liver and kidney.

Functions

1. Copper forms integral part of certain enzymes like cytochrome oxidase and catalase.

2. Ceruloplasmin a copper containing protein is required for iron absorption.

3. Superoxide dismutase (SOD) contains copper ion, SOD converts super oxide radicals to hydrogen peroxide.

4. It is also present in cytochrome C, which is involved in electron transport in mitochondria.

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